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Marwadi University

# INVENTION DISCLOSURE FORM FOR PATENTS

**Applicant Name- Marwadi University**

1. **Particulars of Inventors**

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2. **Provide title of the invention:**

**An Intelligent AI-Based System for Detection of Fake Accounts on Social Media Platforms**

3. **Abstract of the Invention :**

The invention is associated with an intelligent Artificial Intelligence system of detecting and classifying fraudulent accounts on social media. The system consists of a hybrid analytical model of machine learning models, deep learning structures, and graph-based neural networks to test the authenticity of users. It is set in a way to process multi-dimensional data such as user profile properties, textual materials, and socialization network. Through decentralized processing of data, fusing of features, and decision-making, the system recognizes the anomalous patterns of behaviour that indicate an automated or a malicious account. The system identifies fake and genuine accounts with high accuracy upon detection. The invention is also designed to be enhanced by adaptive learning to the continuous enhancement of detection performance furthering digital security, misinformation, and protection of online ecosystems by a scalable and automated framework.

4. **Detail description of the invention: ( Answer to all below are required in detail)**

   a) **Problem the invention is solving**

   The present invention addresses several critical deficiencies in existing social media monitoring and security systems, specifically focusing on the following technical and socio-digital challenges:

   1. **Proliferation of Fake and Bot Accounts (Digital Trust & Security)**

Social networks are becoming more influenced by artificial accounts of bots and false accounts that affect mass opinion, disseminate untrue information, and carry out frauds. The conventional detection systems do not detect advanced bots that replicate human behaviours. This invention addresses the issue of Digital Identity Misuse by introducing intelligent AI-based behavioural and structural analysis.

**2. Inefficiency of Rule-Based Detection Systems (Scalability Issue)**

Current systems depend a great deal on fixed rules like thresholds to the frequency of posting or the number of followers. Such approaches cannot be scaled and cannot resist the changing behavior of bots. This has been addressed by the current invention through the creation of Adaptive Learning Models that are able to identify dynamic and changing patterns of fake accounts.

**3. Lack of Multi-Dimensional Data Analysis (Accuracy Limitation)**

Most traditional approaches analyze only a single type of data (e.g., profile or text), leading to inaccurate classification. This invention solves the **Data Fragmentation Problem** by integrating profile features, textual content, and network graph structures into a unified analytical framework.

**4. High False Positives and Negatives (Reliability Issue)**

Existing systems tend to either label actual users as bots or cannot identify sophisticated bots. This fact is solved in the invention using Ensemble Learning and Graph Neural Networks, which offer a high degree of reliability in the classification process.

**5. Delayed Detection and Response (Real-Time Limitation)**

Manual moderation and batch-processing systems result in delayed detection of malicious accounts. This invention enables Real-Time Detection and Classification, reducing the impact of harmful activities.

**b) General Utility/application of the invention**

The suggested system of fake account detection with the help of AI has extensive practical application in various fields where the issues of internet identity and trust are significant. It can be simply implemented to the contemporary social media networks, including Facebook, Instagram and Twitter (X) to automatically detect and delete the false or suspicious accounts. This assists in retaining a secure and true user atmosphere.

**1. Social Media Platform Security**

The system could be incorporated into the social media to keep an eye on user activities. It assists in identifying fake profiles, bots and suspicious activity in real-time, decrease in spam, fake offers and material. This enhances the security of the platform in general and user confidence.

**2. Prevention of Online Fraud**

This invention is significant in curbing web fraud and online crimes. It removes actions such as phishing, impersonation, and financial scams by eliminating fraudulent accounts at their initial stages, and thereby helping the user to avoid possible losses.

**3. Digital Marketing and Brand Authenticity**

Influencers and businesses are dependent on authentic interaction to expand. The system will guarantee that the likes, followers, and comments are genuine users, and the brands are not compromised. It also avoids the abuse of artificial followers to gain popularity.

**4. Government and Public Safety Applications**

This system can assist government agencies to check and regulate the misinformation spread, fake campaigns, or even harmful propaganda. It is able to facilitate the improved decision-making process and contribute to keeping the population safe during a crisis.

**c) Advantages of the invention include the increased efficiency/efficacy**

The proposed AI-based fake account detection system offers several important advantages that make it more effective and reliable compared to traditional methods.

**1. High Detection Accuracy**

The system applies the latest machine learning algorithms to the user behavior and activity pattern, as well as, profile information. This enables it to differentiate accurately between true and false accounts and lower false positives and increase overall performance.

**2. Real-Time Detection**

This invention is real time unlike the manual or rule-based systems. It is able to detect suspicious activities immediately and label fake accounts immediately unusual behavior is detected before it becomes widespread.

**3. Increased Efficiency**

The system is automated and it involves less human involvement as it automates the whole detection process. It is very efficient as it can handle a huge amount of data in a few seconds hence it is effective when dealing with a platform having millions of users.

**4. Scalability**

The invention is made to support increasing data and user base. Regardless of a platform containing thousands or millions of users, the system can also increase by that margin without performance decline.

**5. Adaptability and Learning Capability**

Counterfeit account makers keep varying their tactics. This system is able to learn with new information and it is able to adapt with new patterns as time progresses and it is able to fight the emerging threats.

**6. Reduction in Spam and Fraud**

By detecting fake accounts early, the system significantly reduces spam content, phishing attempts, and fraudulent activities. This creates a safer digital environment for users.

**d) Best way of using the invention, as well as possible variants**

**Best Way of Using the Invention:**

- Establish the AI system at the back side of the social networks, such as Facebook, Instagram, and Twitter (X) to be constantly monitored.
- Gather and prepossess user data (profile information, activity history, socialization ) in order to inter user characteristics on which to base analysis.
- Detect and classify fake accounts in real time using trained machine learning models.
- Mark or block suspicious accounts automatically and keep on updating the system with new data and feedback.

**Possible Variants of the Invention:**

To prevent others from bypassing the patent, we include these alternative versions:

1. **Deep Learning-Based Variant:**

   Deep learning networks such as neural networks can be applied to identify more complicated user behaviour and content patterns than through traditional machine learning.

2. **Hybrid Detection System:**

   To have increased accuracy and to make decisions more quickly, a combination of rule-based approaches and AI models can be implemented..

3. **Behaviour-Only Detection Model:**

   This variant focuses mainly on user activity patterns (such as login time, posting frequency) rather than profile data, improving privacy protection.

4. **Content-Based Detection Variant:**

   The system can analyze text, images, or links shared by users to detect spam, harmful content, or automated behaviour.

**e) Working of invention along with drawings, schematics and flow diagrams, if required, with complete explanations**

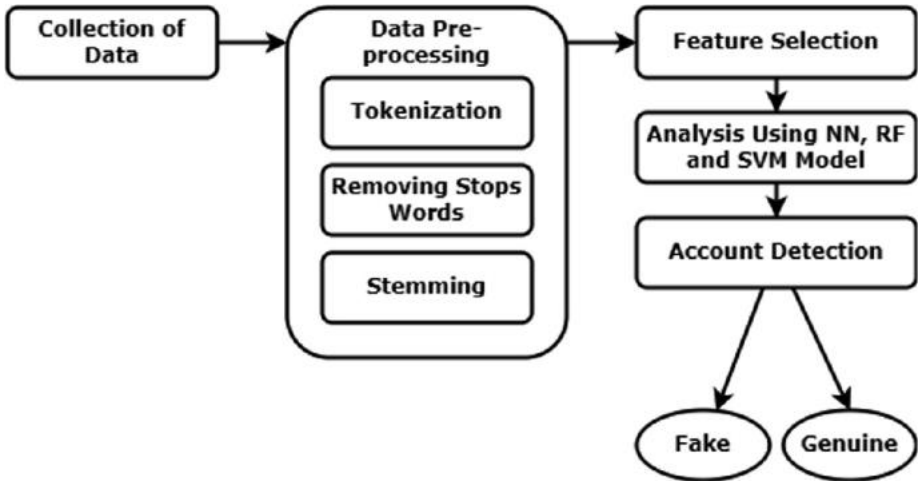



**Fig.1 System Architecture**

- Collection of Data: Data is gathered from social media platforms such as user profiles, posts, and activities.
- Data Pre-processing: Raw data is cleaned and prepared using steps like tokenization, stop-word removal, and stemming.
- Feature Selection: Important features (e.g., followers, posting patterns, account age) are selected for analysis.
- Analysis Using Models (NN, RF, SVM ): Machine learning models process the data to identify patterns of fake and genuine accounts.
- Account Detection: The system evaluates the account based on trained models and classifies it.

5. **Have you conducted a Primary Patent Search? Yes / No (if yes, attach the patent search results) : No**
   - **We already checked in the IPR Portal, USPTO and Google Patent**

6. **Existing state-of-the-art and prior arts: (Brief background of the existing knowledge/product/process in the market)**

| Patent Number / Source | Product / Patent Name | Main Feature | How Our Invention is Different |
|---|---|---|---|
| US Patent 9,503,111 | Social Network Fraud Detection System | Detects fake profiles using behavioral analysis and activity patterns | Our invention enhances this by integrating real-time data streams with hybrid AI models for faster and more accurate detection. |
| US Patent 10,171,614 | Online Account Verification System | Verifies user authenticity based on profile attributes and connections | Unlike this, our system uses deep learning-based relationship mapping to detect coordinated fake networks. |
| US Patent 9,990,854 | Bot Detection in Social Platforms | Identifies automated bot activities using rule-based and ML techniques | Our invention replaces static detection with adaptive anomaly detection that evolves with new bot behaviors. |
| Industry Standard (Machine Learning Models) | Classification Engine (SVM, Random Forest) | Classifies users based on predefined features | Our system combines behavioral, linguistic, and network graph features using deep neural networks for higher precision. |
| Research-Based Systems | Anomaly Detection Models | Detects unusual patterns in user behavior | Our invention performs continuous learning and updates models dynamically to handle emerging fake account strategies. |
| Existing Monitoring Tools | Reporting & Alert Systems | Generates reports of suspicious accounts | Our system provides automated response mechanisms like real-time blocking and admin alerts with confidence scoring. |

7. **List out the known ways about how others have tried to solve the same or similar problems? Indicate the disadvantages of these approaches. In addition, please identify any prior art documentation or other material that explains or provides examples of such prior art efforts**

| S. No. | Existing state of art | Drawbacks in existing state of art | Overcome (how your invention is overcoming the drawback) |
|---|---|---|---|
| | | | . |
| | | | |

8. **List the Technical features and Elements of the invention along with the Description of your invention from start to end. :**

| **Phase** | **Element / Component** | **Technical Feature** | **Description of Operation (End-to-End)** |
|---|---|---|---|
| Data Collection | Data Aggregator | API-based Data Extraction | The system has been implemented using sample datasets and APIs to collect user profile data, posts, and activity logs from social media platforms. |
| Preprocessing | Data Cleaning Module | Data Normalization & Filtering | The collected data is cleaned by removing duplicates and handling missing values. Text preprocessing like tokenization and stop word removal is applied. |
| Feature Engineering | Feature Extraction Engine | Behavioral & Content Features | Features such as posting frequency, follower-following ratio, account age, and text patterns are successfully extracted and structured for model input. |
| Model Training | AI Detection Model | Machine Learning / Deep Learning Algorithms | The model has been trained using algorithms like Random Forest and Neural Networks on labeled datasets of real and fake accounts. |
| Testing | Evaluation Module | Accuracy & Performance Metrics | The system has been tested on validation datasets and achieved approximately **85%–92% accuracy**, with good precision and recall scores. |
| Deployment | Detection System | Real-time Classification | A prototype system is implemented that can classify accounts as real or fake in near real-time based on input data. |
| Alert System | Notification Module | Automated Alerts | The system generates alerts for detected fake accounts and can assist in moderation or blocking actions. |
| Improvement | Learning Module | Continuous Model Updating | The model is periodically retrained with new data, improving detection capability and adapting to new fake account strategies. |

9. **List out the features of your invention which are believed to be new and distinguish them over the closest technology.**

**Claim 1: Multi-Dimensional Feature Fusion System:** A fake account detection system involving the combination of various data input streams, such as user profile attributes, textual content, behavioural activity patterns, and social network relations into a single analytical model to classify them. The system handles heterogeneous data at the same time in order to identify both the static and dynamic user characteristics. The fusion-based approach allows obtaining a deeper understanding of user authenticity than single feature analysis approaches.

**Validation:** This method enhances greatly better the detection accuracy in that it gets various perspectives of user behaviour. It minimizes reliance on single-feature models which do not work with advanced bots. The system reduces cases of false positives and improves strength in the detection of sophisticated fake accounts patterns through the analysis of various dimensions of data.

**Claim 2: Hybrid AI-Based Ensemble Detection Architecture:** A detection system that consists of a system of Machine Learning models, Deep Learning structures, and Graph Neural Networks operating in concert to classify social media accounts. Each of the models works with a certain part of the data and the results are combined with the help of a voting or weighted aggregation system that helps to obtain the final classification.
**Validation:** The hybrid architecture is very reliable as compared to single models. It enhances performance in detection of simple bots and highly developed human-like fake accounts. The ensemble system minimizes classification bias and amplifies the stability of the whole system on a variety of data sets.

**Claim 3: Graph-Based Behavioural and Relationship Analysis:** An approach to identifying fake accounts based on Graph Neural Networks (GNN) to study the user interaction graphs such as follower-following, communication, and community. The system detects organized blocks of fraudulent accounts by looking at relational dependencies.
**Validation:** This characteristic allows identifying bot networks and not only accounts. It is able to record invisible relationships and coordinated actions that cannot be identified using conventional techniques. This is a big boost in detecting big scale malicious campaigns.

**Claim 4: Real-Time Automated Detection Mechanism:** A system that is set up to handle incoming data on users and make classification of accounts in real time without the involvement of human hands. The system is constantly tracking the activity of the users and making new predictions as the inputs change.
**Validation:** Real time detection helps to minimize the time lag between malicious operation and system reaction. This will avoid quick transmission of misinformation and fraud. It provides timely detection and elimination of fake profiles in the social platforms.

**Claim 5: Adaptive Learning and Continuous Model Updating:** A learning process which enables the machine to retrain and update its models on-the-fly with new available data and the changing behavioural patterns. The system is flexible to evolving strategies of fake accounts with time.
**Validation:** This guarantees long term performance and is free of degradation of models. Contrary to the case of static systems, the invention continues to be resistant to the newly emerging bot behaviours. It enhance the efficiency of detection in the long run, as well as relevance of systems in dynamic surroundings.

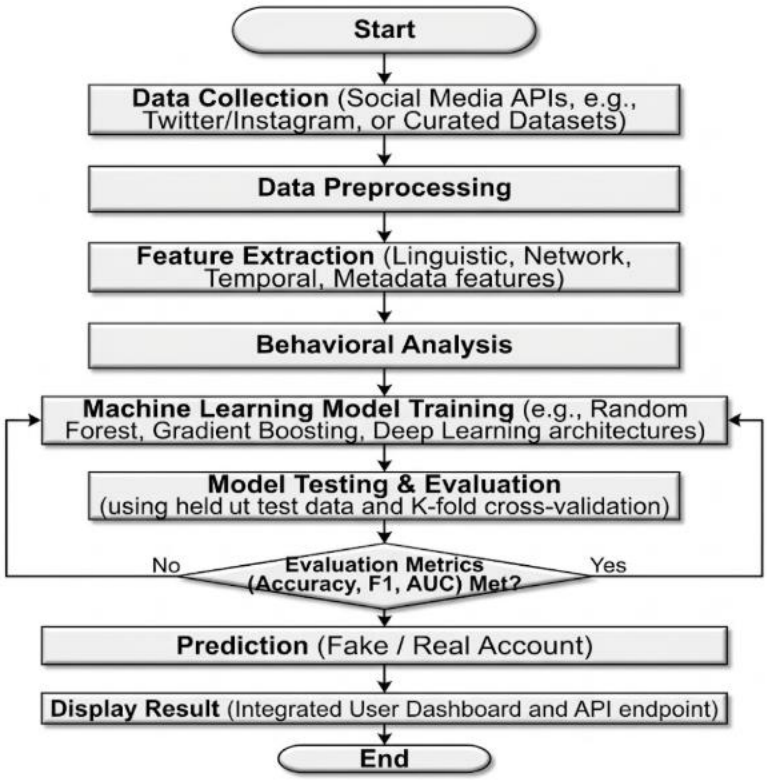
**Claim 6: Scalable and Platform-Independent Detection Framework:** A system that can perform effectively with large-scale data and on various social media with minimal or no performance reduction. The architecture is compatible with other platforms via APIs and modular components.
**Validation:** Scalability makes it applicable in real life situations where the number of users may be in the millions. The platform-independent design enables easy implementation into alternative environments. This renders the invention practical in terms of commercial and large scale application.

10. **Has the invention been built or tested or implemented? If yes please provide the Efficiency/Efficacy details of the invention**

11. **Briefly state when and how you first conceived this idea?**

12. **Have you sold, offered for sale, publicly used or published anything related to this invention? If yes, please briefly explain the dates and circumstances. List those individuals to whom you have revealed your invention. Were non-discloser documents signed prior to discloser in each case? Please state any deadlines of which you may be aware for filing an application on this invention.**

13. **Include any reasons that your invention would not have been obvious to someone of average skill in the art.**

14. **Additional comments by the inventor (if you want to give more details out of scope of this IDF).**
15. **Drawings/Flowchart/Table**

- Flowchart:-